

Private Access (ZPA)

Configuring a Log Receiver

A log receiver is a storage location that can receive information about App Connectors and users. Your App Connectors must be deployed prior to configuring a log receiver. To learn more, see the [App Connector Deployment Guides for Supported Platforms](#).

Logs go to the App Connector group that is geographically closest to the instance of the Log Streaming Service (LSS). In the event that the App Connectors in a previously selected App Connector group are not available, then log transmission switches to App Connectors in another location. If each App Connector group points to a different log receiver and is configured for the exact same LSS, then the same log is sent to an App Connector in each App Connector group. The LSS attempts to resend a log if it determines that a log has not been received by the SIEM system.

Adding a Log Receiver

To add a log receiver:

1. Go to Logs > Log Streaming > Log Receivers.
2. Click Add.

The Add Log Receiver window appears.

3. In the Add Log Receiver window, configure the following tabs:

▼ 1. [Log Receiver](#)

- a. On the Log Receiver tab:

- **Name:** Enter a name for the log receiver. The name cannot contain special characters, with the exception of periods (.), hyphens (-), and underscores (_).
- **Description:** (Optional) Enter a description.
- **Domain or IP Address:** Enter the fully qualified domain name (FQDN) or IP address for the log receiver.

If the FQDN or IP address of the log receiver overlaps with or is the same as the wildcard domain or IP subnet defined in an application segment, the [Bypass setting](#) configured for the application segment takes precedence. As a result, if the FQDN or IP address is bypassed for a user on a trusted network, the user's device cannot communicate with the log receiver.

- **TCP Port:** Enter the TCP port number used by the log receiver.
- **TLS Encryption:** Enable to encrypt traffic between the App Connector and the log receiver using mutually authenticated TLS communication. To use this setting, the log receiver must support TLS communication. To learn more, see [About the Log Streaming Service](#). By default, this setting is disabled.
- **App Connector Groups:** Choose the App Connector groups that can forward logs to the receiver. You can search for a specific group, click **Select All Displayed** to apply all groups, click the **Delete** icon next to a selection to remove it, or click **Clear All** to remove all selections.



If you have a use case where the user's device needs to send logs to the log receiver using Private Access, configure an application segment with the log receiver domain or IP address and the port that the log receiver is listening on.

b. Click Next.

Add Log Receiver



Log Receiver > Log Stream > Review

Name

Description

Domain or IP Address

TCP Port

TLS Encryption

Enabled

☒ Disabled

App Connector Groups

Choose one or more App Connector Groups

Next

Previous

Cancel

2. Log Stream

a. On the Log Stream tab:

i. Log Type: Select an option from the drop-down menu:

- **User Activity:** Information on end user requests to applications. To learn more, see [Understanding User Activity Log Fields](#).
- **User Status:** Information related to an end user's availability and connection to Private Access. To learn more, see [Understanding User Status Log Fields](#).
- **App Connector Status:** Information related to an App Connector's availability and connection to Private Access. To learn more, see [Understanding App](#)

Connector Status Log Fields.

- **Private Service Edge Status:** Information related to a Private Service Edge's availability and connection to Private Access. To learn more, see [Understanding Private Service Edge Status Log Fields](#).
- **Private Cloud Controller Status:** Information related to a Private Cloud Controller's availability and connection to Private Access. To learn more, see [Understanding Private Cloud Controller Status Log Fields](#).
- **Browser Access:** HTTP log information related to Browser Access. To learn more, see [Understanding Browser Access Log Fields](#) and [About Browser Access](#).
- **Audit Logs:** Session information for all admins accessing the Zscaler Admin Console. To learn more, see [Understanding Audit Log Fields](#) and [About Audit Logs](#).
- **AppProtection:** Information related to AppProtection policy activity in your organization. To learn more, see [Understanding AppProtection Log Fields](#).
- **App Connector Metrics:** Information related to an App Connector's metrics. To learn more, see [Understanding App Connector Metrics Log Fields](#).
- **Private Service Edge Metrics:** Information related to a Private Service Edge's metrics. To learn more, see [Understanding Private Service Edge Metrics Log Fields](#).
- **Private Cloud Controller Metrics:** Information related to a Private Cloud Controller's metrics. To learn more, see [Understanding Private Cloud Controller Metrics Log Fields](#).
- **Microsegmentation Flow Logs:** Information related to Microsegmentation flow logs. To learn more, see [Understanding Microsegmentation Flow Log Fields](#).
- **KRB Inspection Logs:** Information related to Kerberos inspection log activity in your organization. To learn more, see [Understanding KRB Inspection Log Fields](#).
- **LDAP Inspection Logs:** Information related to LDAP inspection log activity in your organization. To learn more, see [Understanding LDAP Inspection Log Fields](#).
- **SMB Inspection Logs:** Information related to SMB inspection log activity in your organization. To learn more, see [Understanding SMB Inspection Log Fields](#).

ii. **Log Template:** Select an option from the drop-down menu:

- **CSV:** The template uses a comma-separated values format.
- **TSV:** The template uses a tab-separated values format.
- **JSON:** The template uses a JavaScript Object Notation format.

iii. **Log Stream Content:** The default content that is displayed changes based on the **Log Type** and **Log Template** you selected in previous steps.

You can also edit the log stream content within the text field in order to capture specific fields and create a **Custom** log template. To learn more, see [Understanding the Log Stream Content Format](#).

If you made changes to the log stream content, the **Restore Default Content** option appears, allowing you to revert to the default.

iv. **Policy:** Define a streaming policy for the log receiver. For example, you can create a policy where the receiver only captures logs for a specific segment group or a specific set of session status error codes. The criteria you can use is dependent upon the **Log Type** you select.

- Under **SAML and SCIM Attributes** or **User and Group Attributes**, click **Select IdP** and choose the IdP configuration you want to include in the policy. The IdP must be configured for User SSO. To learn more, see [Configuring an IdP for Single Sign-On](#). If you need to include multiple IdPs in the policy, click **Select IdP** again. Click **Select SAML and SCIM Attributes** to add the following attributes to the policy:
 - **SAML Attributes** or **Session Attributes:** Click **Any SAML Attribute** if you want to allow logs for any user to be streamed (i.e., the policy applies to any users, groups, departments, etc.). By default, the policy is set to **Any SAML attribute from an IdP**. Click a specific [SAML attribute](#) if you want to allow logs for only specific users, groups, departments, etc. to be streamed. You can search for a specific group, click **Select All Displayed** to apply all groups, click the **Delete** icon next to a selection to remove it, or click **Clear All** to remove all selections. After you make a selection, enter the SAML attribute value (i.e., the users to whom the policy applies) in the text field that appears. Click **Add More** to add multiple attributes, if necessary.
 - **SCIM Attributes** or **User Attributes:** Click **Select a SCIM attribute name** and select the SCIM attribute you want to be streamed. You can also search for a specific SCIM attribute. Click **Choose SCIM User** and select the SCIM User. Click **Add More** to add multiple SCIM attributes, if necessary.

- **SCIM Groups Attributes or Group Attributes:** Click **Select a SCIM group name** and select the SCIM group you want to be streamed. You can also search for a specific SCIM group. Click **Add More** to add multiple SCIM groups, if necessary.

If you are subscribed to ZIdentity and have this feature enabled for your tenant, the **SAML and SCIM Attributes** option is replaced with **User and Group Attributes**. To learn more, see [What Is ZIdentity?](#)

If you added multiple SAML attributes to the policy, an OR Boolean operator is used between them by default. For example, if you selected **FirstName** and **LastName**, the policy is only applied to users with the specified **FirstName** OR **LastName** for that IdP. However, you can toggle this to an AND operator by clicking on it.

POLICY

SAML and SCIM Attributes Application Segments Segment Groups Client Types Connection: Status Code

SAML and SCIM Attributes

↳ IDP: Example IdP

↳ SAML Attributes

Select IdP

Select SAML and SCIM Criteria

firstName_scim_saml = Jane

email_scim_saml = example@safemarch.com

<OR>

Add More

If you selected multiple IdPs for the policy, an OR Boolean operator is used between them by default. For example, you can select one IdP that includes **FirstName** OR **LastName**, and another IdP that includes **Any SAML Attribute**. In this case, the policy applies to a user authenticating from the first IdP if they have the specified **FirstName** OR **LastName**, *or* it applies to any user authenticating from the second IdP.

POLICY

[SAML and SCIM Attributes](#) [Application Segments](#) [Segment Groups](#) [Client Types](#) [Connection: Status Code](#)

SAML and SCIM Attributes + Select IdP

↳ IDP: TheZeroTrustExchange Okta + Select SAML and SCIM Criteria

↳ SAML Attributes

Any SAML attribute + ▼

+ Add More

<OR>

↳ IDP: TheZeroTrustLab Entra ID + Select SAML and SCIM Criteria

↳ SCIM Attributes

Last Name + ▼ = Sherman + ▼

+ Add More

If your [IdP configuration for SSO](#) includes SAML attributes from  multiple IdPs, Zscaler recommends that you do not use the AND Boolean operator.

- **Application Segments:** Choose the [application segments](#) you want to include from the drop-down menu. You can search for a specific application segment, click **Select All Displayed** to apply all application segments, click the **Delete** icon next to a selection to remove it, or click **Clear All** to remove all selections. If you don't make any selections, then the policy applies to any application segment.

 There are limits to the number of application segments applied to a rule. For a complete list of ranges and limitations for Log Receiver Policy rules, see [Ranges & Limitations](#).

- **Segment Groups:** Choose the [segment groups](#) you want to include from the drop-down menu. You can search for a specific segment group, click **Select All Displayed** to apply all segment groups, click the **Delete** icon next to a selection to remove it, or click **Clear All** to remove all selections. If you don't make any selections, then the policy applies to any segment group.
- **Client Types:** Choose the client types (i.e., **Web Browser**, **Machine Tunnel**, **Cloud Connector**, **Client Connector for VDI**, **Client Connector**, **Extranet**, **ZPA LSS**, **ZIA Service Edge**, **Client Connector Partner**, and **Branch Connector**) you want to include from the drop-down menu. You can search for a specific

client type, click **Select All Displayed** to apply all client types, click the **Delete** icon next to a selection to remove it, or click **Clear All** to remove all selections. To learn more, see [What Is Zscaler Client Connector?](#), [About Browser Access](#), [About Machine Tunnels](#), [About Extranet](#), [About Cloud Connectors](#), [About Branch Connectors](#), and [What Is Zscaler Client Connector for VDI?](#) If you don't make any selections, then the policy applies to any client type.

- **Connection: Status Code:** Choose the [session status codes](#) you want to exclude from the drop-down menu. You can search for a specific status code, click **Select All Displayed** to apply all status codes, click the **Delete** icon next to a selection to remove it, or click **Clear All** to remove all selections. If you don't make any selections, then the policy includes any status code.

 With the exception of session statuses, all of these fields require configuration to *include* the values. Session Statuses are configured to *exclude* values.

If you define multiple criteria for the streaming policy, an OR Boolean operation is assumed between Application Segments and Segment Groups, and an AND operation is assumed between SAML Attributes, Application Segments or Segment Groups, Client Types, and Session Statuses. This means that SAML Attributes AND (Application Segments OR Segment Groups) AND Client Types AND Session Statuses logic is used for the policy.

For example, if you specify a SAML attribute, an application segment, and a session status code, then LSS only streams information whenever that attribute's value and application segment appear in the log, and the specific status code does not appear.

When streaming a log, a Private Access LSS client is seen by the service as a user who is authenticating and sending requests to the log receiver. This user represents the LSS service, not an actual user. However, a User Status and User Activity log is generated corresponding to the Private Access LSS client.

 In the streaming policy when **SAML Attributes** is set to **Any SAML Attribute**, it is interpreted by Private Access to include all users, including the Private Access LSS client. If you do not want logs from the Adaptive Access Engine LSS client to stream to your log receiver, you can modify the streaming policy to do one of the following:

- Specify the application segments that must have logs streamed.
- Specify the segment groups that must have logs streamed.
- Specify SAML attributes instead of using the allow any attribute default.

b. Click Next.

Add Log Receiver ✕

[Log Receiver](#) > [Log Stream](#) > [Review](#)

CONFIGURATION

Log Type

User Activity

Log Template

CSV

Log Stream Content

```
%s{LogTimestamp:time} User Activity zpa-Iss:
,%s{Customer},%s{SessionID},"%s{ConnectionID}",%s{InternalReason},%s{ConnectionStatus},%d{IPProtocol}
,%d{DoubleEncryption},%s{Username},%d{ServicePort},%s{ClientPublicIP},%s{ClientPrivateIP},%f{ClientLatit
ude},%f{ClientLongitude},%s{ClientCountryCode},%s{ClientZEN},%s{Policy},%s{Connector},%s{ConnectorZ
EN},%s{ConnectorIP},%d{ConnectorPort},%s{Host},%s{Application},%s{AppGroup},%s{Server},%s{ServerIP},
%d{ServerPort},%d{PolicyProcessingTime},%d{ServerSetupTime},%s{TimestampConnectionStart:iso8601},
```

POLICY

SAML and SCIM
Attributes

[Application
Segments](#)

[Segment
Groups](#)

[Client
Types](#)

[Connection: Status
Code](#)

SAML and SCIM Attributes

[+ Select IdP](#)

Next

[Previous](#)

[Cancel](#)

3. Review

On the Review tab, review your log receiver configuration, and click **Save**.

X

Add Log Receiver

Log Receiver > Log Stream > Review

Log Receiver Example Log Receiver	Domain or IP Address:Server Port:TCP myexample.com:80:TCP	TLS Encryption ⛔ Disabled
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App Connector Groups

Default Group

Log Type User Activity	Log Template CSV
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SAML and SCIM Attributes

IDP: Example IdP

↳ SAML Attributes

firstName_scim_saml = Jane OR email_scim_saml = example@safemarch.com

Review all of the information before clicking Save

Save

Previous

Cancel

Copying a Log Receiver

To copy a log receiver:

1. Go to Logs > Log Streaming > Log Receivers.
2. In the table, locate the log receiver you want to copy and click the Copy icon.

The Add Log Receiver window appears.

3. In the Add Log Receiver window, modify fields as necessary.
4. Click Save.

Editing a Log Reciever

To edit a log receiver:

1. Go to Logs > Log Streaming > Log Receivers.
2. In the table, locate the log receiver you want to modify and click the Edit icon.

The Edit Log Receiver window appears.

3. In the Edit Log Receiver window, modify fields as necessary.

4. Click **Save**.

Deleting a Log Receiver

To delete a log receiver:

1. Go to **Logs > Log Streaming > Log Receivers**.
2. In the table, locate the log receiver you want to delete and click the **Delete** icon.
3. In the confirmation window that appears, click **Delete**.